

# MCQS Final Term Spring 2024

## Bio505 Current Quizzes For Final Term Spring 2024

By Muhammad Zaman

Zoology Legend TEAM

Final Term Spring 2024

1. Which is not Autosomal dominant disease? **Gaucher's disease**
2. In ZW-ZZ system, which is male? **ZZ is male**
3. Down syndrome is due to trisomy of chromosome **21**.
4. Cell Lysis Buffer is **non-ionic detergent**, designed to lyse outer cell membrane of blood and epithelial cells, but will not break down nuclear membrane.
5. Darker bands in R staining are **Euchromatin**.
6. Annealing temperature in PCR **40-68°C**.
7. **Q-PCR** is commonly used to determine whether a DNA sequence is present in a sample and the number of its copies in the sample.
8. Allele-specific PCR used for identifying of **SNPs**.
9. Whole genomes can be amplified by **Whole genome PCR**.
10. Mini Primer PCR can extend from **short primers**.
11. Northern blot is a method used for detection of a **specific RNA sequence**.
12. Which describes best the prokaryotic chromosomes? **Single, circular DNA molecule**.

13. Prokaryotic nucleus lies in **nucleoid**.
14. **Pedigrees** are used to determine the pattern of inheritance of a genetic disease within a family.
15. Variations are due to **Genes & environment**.
16. In ring chromosome, **two arms joined** joins to form ring shape.
17. Human zygote is diploid and **\*46 chromosomes**.
18. **H1 Histone** tightly pack the nucleosome.
19. Z-DNA is **left-handed helix**.
20. Which is Transition mutation? Purine is replaced by purine. Or Pyrimidine is replaced by pyrimidine. **Both are correct**.
21. Growth takes place by **Mitosis**.
22. **Non-kinetochore** spindle fibers push against each other that stretch the cell into an oval shape in **Anaphase**.
23. Meiosis I results in **two haploid daughter** cells.
24. Cystic fibrosis and muscular dystrophy are **Single gene disorders**.
25. Genes switched on when have **unmethylated cytosine**.
26. Trisomy is **Aneuploidy**.
27. Missense mutation produces a change in amino acid sequence in **protein product**.
28. Which is not true about Autosomal recessive disease? **Diseases occurs in individuals with two mutant alleles**. This one is not true.
29. Homozygotes with two copies of the altered gene. **It's true**.
30. Frequency of autosomal dominant disorders is **highest**.
31. Microbiology, DNA fingerprinting, Genetic diversity, Forensics, Mapping of genomes are applications of? **VNTRs**.
32. Male pattern baldness is a **recessive sex-linked trait** on the **X chromosome**.

33. A woman, whose father had male pattern baldness, marries a man with this trait. What is the probability that any son born will have pattern baldness? The probability that a son will be born with pattern baldness is **50%**.
34. Both a husband and wife have normal vision but their son is color blind. What are the likely genotypes of the boy's parents? **Boys Recessive**
35. Blood cells are lysed **erythrocytes lysis and leukocytes lysis**.
36. Dissolve RNA pellet by adding **RNAse-free H<sub>2</sub>O**.
37. In PCR, DNA is **amplified**.
38. About **25000 SNPs** in humans are ... **non- synonymous** .
39. BP wrapped around them. (**8, and 147** ).
40. Annealing temperature..... **40-68°C** .
41. In organic method of DNA purification is used to remove the protein. \_(**phenol chloroform**).
42. ▪ In case of X linked recessive disorder A Cross b/w abnormal and normal female Will be ... (**50%** ).
43. RNA is present in (**few viruses**)
44. nucleosome is composed of a core of eight histone proteins and DNA (**147 bp**)
45. Flemming discovered ... (**Lambrush** ).
46. Chance of inheriting a mutant allele from one parent is 50% and for the other parent is also 50%. Net chance of inheritance of two mutant alleles is ....**25%**.
47. Symbol of substitution \_\_ **">"** .
48. A mutation that are responsible for shorten of protein due to producing stop codons....**non sense mutation**
49. Fur color in labrador retriever controlled by....**2gene** .

50. **Q-PCR** is commonly used to determine whether a DNA sequence is present in a sample and the number of its copies in the sample
51. How many cases of Turner syndrome are due to ring chromosome. **Ans 20%**
52. Dark region rich in .... (A/T) and light in..... **(G/C)**
53. RNA is genetic material in.....**some viruses**
54. nucleosome is composed of a core of eight histone proteins and DNA ( **147 bp**) wrapped around them.
55. During RNA purification, RNA pellet is dissolved by adding.....to each sample.  
**RNAse-free H<sub>2</sub>O**
56. In standard PCR procedure the required annealing temperature is In mutation a single base pair is altered..... **40-68°C**
57. Major ingredients of Polymerase Chain Reaction (PCR) is/are:.....  
**buffer, MgCl<sub>2</sub>, primers, DNA template (All)**
58. Mutation at lower case, **RNA level**
59. The purpose of southern blot test is to .....**DNA**
60. Mt DNA genes in human **37**
61. Mini Primer PCR can extend from \_\_**short primers.**
62. Rabbit having light grey colour\_\_**chinchilla**
63. Blood cells are lysed \_\_\_\_**erythrocytes lysis**
64. Which of the following is not an example of multi-locus marker \_\_\_\_**SST**
65. Northern blotting is used to for \_ **RNA**
66. During RNA purification, RNA pellet is dissolved by adding.....to each sample.  
**RNAse-free H<sub>2</sub>O**
67. "\_" (underscore) indicates a range of affected residues, separating the first and last residue affected
68. Amniotic fluid analysis to determine genetic and chromosomal disorders after .....**14 weeks gestation.**

69. Microbiology, DNA fingerprinting, Genetic diversity, Forensics, Mapping of genomes are applications of? \_\_\_ **VNTRs**.
70. Which one of the statement is correct about colour blindness is \_\_\_ **Affected individual** can't differentiate the Green and red colour
71. Chromosomes have centromere at one end of the arms ... **telocentric**
72. .... is a chelating agent. \_\_\_ **EDTA**
73. ☐ **PCR** is a technique used for nucleic acid...
74. Southern blot is use for \_\_\_ **DNA**
75. Pattern baldness is a recessive trait. A female inherits it from her father and marries a man who has this trait . What are the chances to pass on this trait **50%**
76. Expression of phenotype by a number of individuals is called as \_\_\_ **phenotypic variation**
77. Geimsa stain is used for \_ chromosomal banding (**C-banding**)
78. Trisomy 18 causes which disease **Edward syndrome**.
79. Who discovered gene map in Drosophilla? **T.H Morgan**
80. How many no. of genes are present in human mitochondrial genome **37**.
81. Which of the following is not an example of multi-locus marker **SST**.
82. Which PCR is used to determine **SNPs Allelic specific PCR**.
83. Dark bands are rich in A.T , light bands are rich in G.C.
84. Whole genomic size & amplification is done by **Whole genome PCR**.
85. Male pattern baldness wala jo lmbaaaaa sa quiz h..answer **50%**.
86. Fur color in Labrador retriever controlled by 2 genes .
87. Annealing temp **40-65**.
88. Y chromosome kis leay HOTA hai (**Maleness**)
89. Smooth curve is property of? **Continuous variation**
90. Large arm of typical chromosome is named as?
91. Due to ring chromosome, % of turner syndrome? **20%**

92. Round worm has only two chromosomes. Protozoa having **300 or more**.  
Fungi have **3-8 chromosomes**. Angiosperms have from **8 – 16**
93. CEN DNA is how many base pair long? **120 base pairs long**
94. At which phase spindle fiber attached to centromere? **Metaphase**
95. No of genes in drosophila? **13700**
96. Name if phenomenon that break the occurrence of linkage?
97. How many variations in VNTR? **(8 to > 50 base pairs)**
98. if both parents are heterozygous dominant disease what will be the chance of child not having that disease? **50%**
99. cri- du- chat syndrome is caused by ... **Monosomy**
100. monosomy represent.....? **Turner syndrome**
- 101.** lampbrush chromosome are found in...? **oocytic nuclei of vertebrates (sharks, amphibians, reptiles and birds)**
102. Fragile X syndrom accepted by....? **200 CGG**
103. Epistasis is a interaction of how many genes..?
104.  $\nless HGNC$  stands for...?
105.  $\nless$ who discovered genetics ..?
106. An aneuploid person is female, but her cells have **two Barr bodies**.
107. Two Barr bodies are due to deactivation of **two X chromosomes**
108.  $\nless$  The Type of cell division important for growth is\_\_\_
109.  $\nless$ northern blot is used for\_\_\_
110.  $\nless$ Which is following is not chromosomal aberration\_\_\_\_\_
111.  $\nless$ .R banding has dark bands of\_\_\_
112.  $\nless$ cancer, infertility are result of \_\_\_
113.  $\nless$  Colchicine to promoter\_\_\_ **polyploidy**

114. ✨Cross for determination of genotype of an individual\_\_
115. ✨Autosomal dominant diseases\_\_
116. ✨\_\_is inherited through ovum not sperm
117. ✨sequence difference compared to a reference
118. **(10 genotype)** for a gene has four alleles.
119. ☞Two babies in a maternity ward have lost their identify bands and there is some confusion about their footprint record.( **Either of the baby belongs to her**).
120. ☞Dissolve RNA pellet by adding (**RNAse free H2O**) yo each sample.
121. ☞Annealing temperature is (**40-65°C.**)
122. ☞Nested pcr (**two Pairs**).
123. ☞Northern blotting used for (**RNA**).
124. ☞the giant chromosomes found in the (**diptera**).

**Muhammad Zaman**

**Admin of Zoology LEGENDs**

**Need your Precious Prayers**